Software Requirements Specification

for

E-DBA

Version 1.0 approved

Prepared by Sibo Qiu s230026131

Yucheng Tong s230026146

Zhijinxiang Zheng s230026222

Manxi Tao s230026145

Yixin Xia s230026171

Boxin Su s230026138

Group A04

2025/2/24

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Revision History

Name	Date	Reason For Changes	Version
Boxin Su	2025.2.26	Create the first version of SRS	1.0
Boxin Su	2025.3.5	Add System Features and User Interface	1.1

1. Introduction

1.1 Purpose

More and more organizations need to share data with other organizations. Registering higher education institutions requires secure data sharing with other schools in a transparent and interoperable manner. This SRS specifies the software requirements for the Education Data Bay Area (E-DBA) system. The E-DBA system is designed to provide a secure and efficient platform for data sharing among registered higher education organizations. The scope of this SRS includes the functional and non-functional requirements of the entire system, covering features such as student identity authentication, thesis sharing, course information sharing, and an optional vault service. The SRS also outlines the system's integration with external payment gateways and interfaces for accessing public and private data.

1.2 Document Conventions

This SRS follows standard software engineering conventions for writing software requirements. Each requirement is specified clearly and unambiguously. Functional and non-functional requirements are categorized separately. Requirements are numbered for easy reference, and each requirement is assigned a priority level. System features are further broken down into detailed requirements, ensuring consistency across the document. All technical terms and acronyms are defined in the glossary section.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for a range of stakeholders, including developers, testers, project managers, and system architects. Additionally, it will be useful for documentation writers and marketing staff involved in communicating the features of the system. The remaining parts of this SRS include the Overall Description, System Features, External Interface Requirements, Other Nonfunctional Requirements. Overall Description is an introduction and description of the features and functions of this project. System Features provides a detailed description of the system functionalities and functional requirement classifications for this project. External Interface Requirements are detailed requirements for each system feature and external interface. Other Nonfunctional Requirements include specific performance and security requirements.

Developers, testers, and documentation writers will need to focus on the system’s functional requirements and detailed specifications. They should begin by reviewing the Overall Description section for a high-level overview, then proceed to the System Features and External Interface Requirements sections for more in-depth technical details. Project managers and marketing staff should begin by reviewing the Overall Description to understand the product’s features, goals, and target audience. They may also find the External Interface Requirements useful in understanding how the system will interact with other tools or platforms. End users will primarily interact with the system’s features and functionalities. They should focus on System Features to understand the main functions they will use, particularly the Data Sharing, Thesis Sharing, and Student Identity Authentication functionalities.

1.4 Project Scope

The E-DBA system is a data-sharing platform designed to facilitate secure data exchanges between higher education institutions. The scope of this project encompasses the creation and deployment of a system that enables institutions to share and access data in a controlled and transparent manner, while ensuring compliance with data sovereignty regulations.

The E-DBA system provides a secure platform for higher education organizations to share data and services. It includes features for verifying student identities, sharing academic theses with configurable access permissions, and enabling the management and sharing of course information. Additionally, it offers an optional data vault service for organizations that cannot maintain their own databases, allowing them to securely store and share data. The system also supports online payment integration, enabling users to access paid services through multiple payment methods, including transfers, PayPal, WeChat, Alipay, and credit cards. The system will be fully integrated with external data sources, such as student record databases and thesis repositories, to ensure smooth interoperability. It will also interface with external payment gateways for processing payments related to data access and service usage.

This SRS covers the requirements for the E-DBA system, including both functional and non-functional requirements, ensuring that all aspects of the system's design, performance, security, and usability are addressed. The goal is to create a scalable, secure, and easy-to-use platform that meets the needs of academic institutions and facilitates the sharing of important academic data.

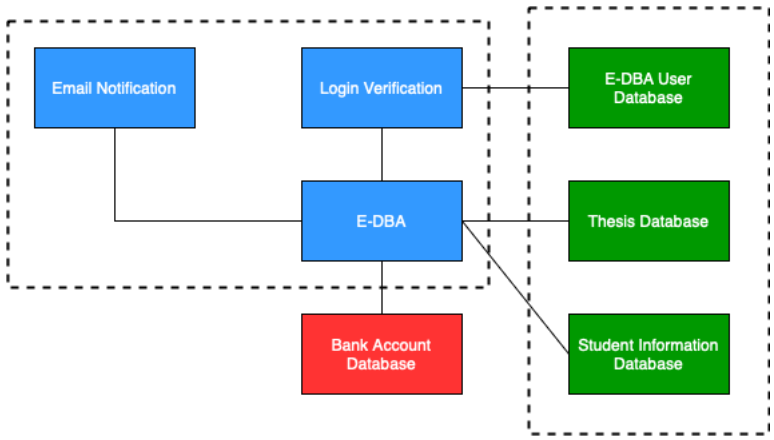
1.5 References

International Dataspace Association (IDS) Dataspace Model. (https://internationaldataspaces.org/)

2. Overall Description

2.1 Product Perspective

The E-DBA is a novel, standalone service system that provides the following services: course information access, thesis sharing, student identity authentication, and student GPA record access. The system consists of a total of 7 modules: E-DBA, Login Verification, Email Notification, E-

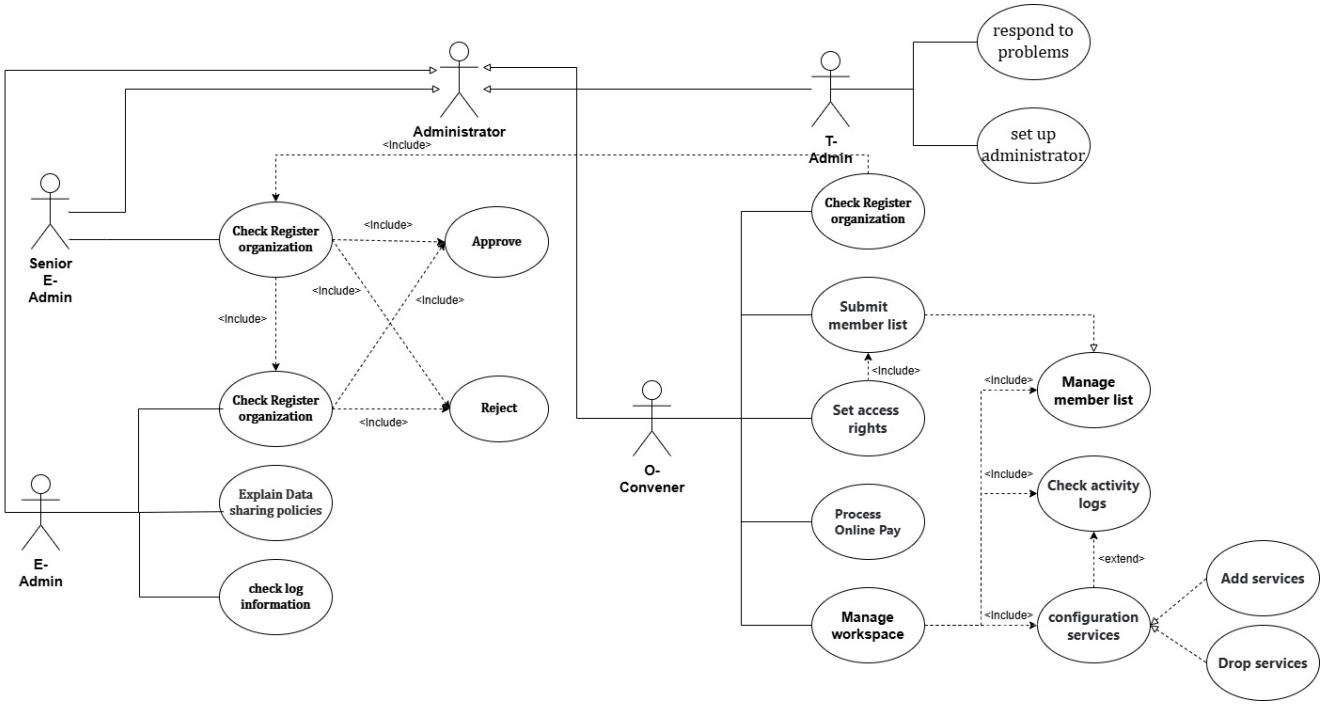


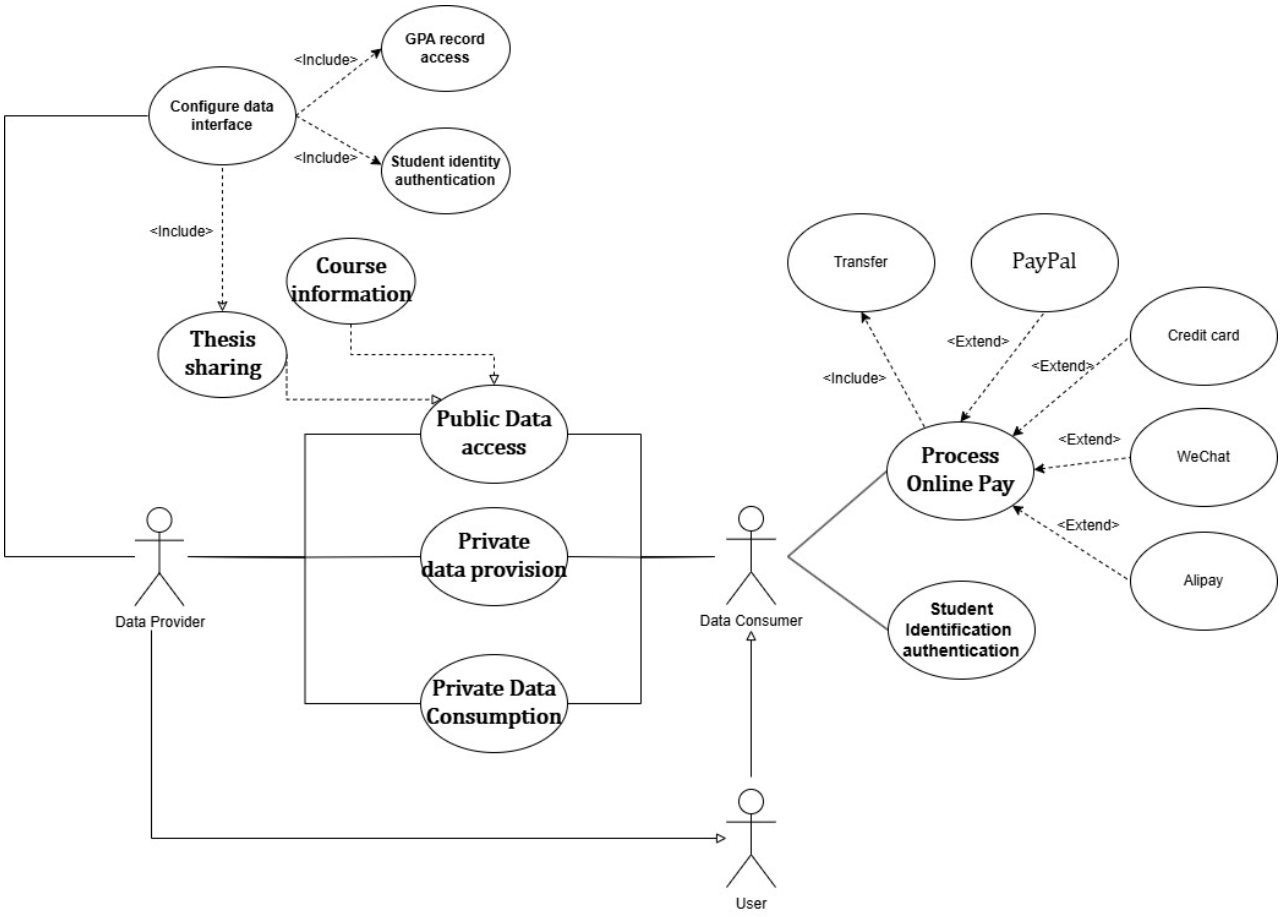
DBA User Database, Student Information Database, Thesis Database, and Bank Account Database. The system overview diagram is as follows:

2.2 Product Features

E-DBA is a data service platform that provides both public and private data access. Public data access allows any user to retrieve course information and use the thesis-sharing service, which may incur charges regardless of the user's organization. Private data access, available within each organization, includes student identity authentication (charged per inquiry) and access to student GPA records, both sharing a similar interface.

The system supports multi-level admin roles: Technical Admin (T-Admin) for system health monitoring and help desk support, Management Admins (E-Admin and Senior E-Admin) for policy enforcement and registration approvals, and O-Conveners for managing member lists and workspace activities. E-DBA also offers online payment processing through multiple methods with automated billing for membership fees and service usage. An optional vault service provides secure data storage for organizations without in-house database infrastructure, featuring configurable access rights and API integration for external analytics. Robust audit and compliance features, such as immutable activity logs and automated policy enforcement for data sovereignty and role-based access control, ensure secure and efficient operations.





O-Convener Registration Scenario

1. **Access Registration Page**: The O-Convener accesses the E-DBA system's registration entrance without logging in.
2. **Submit Application**: Fills in personal and organization details, uploads required proof docs, enters a valid email, agrees to terms, and clicks "Submit".
3. **System Check**: The system verifies filled fields, file formats, and email. If there are errors, it prompts the O-Convener to correct and resubmit.
4. **Enter Review Queue**: After passing the initial check, the application gets a unique ID, is stored in the database, and an email is sent to confirm submission. It then waits for E-Admin's review.
5. **E-Admin Review (Approval Assumed)**: The E-Admin checks the application. If approved, it goes to the Senior E-Admin; if not, the O-Convener is notified with reasons.
6. **Senior E-Admin Approval**: The Senior E-Admin reviews and, if satisfied, approves, and the system creates a workspace for the organization.
7. **Login Setup**: The O-Convener receives a login link and temp password via email, logs in, and sets a new password.
8. **Manage Organization**: Enters the workspace, can complete organization info (optional), and adds members by file upload or manual input. The system validates member info, and after that, the O-Convener can use system functions.

2.3 User Classes and Characteristics

Users are divided into two major types: Administrators and Data Consumers/Providers.
For administrators, there are three levels: T-Admin, E-Admin, and O-Convener:

- **T-Admins** are responsible for handling requests from other users and setting up the two management administrator roles.
- **E-Admins** are tasked with inspecting registration applications and explaining data-sharing policies.
- **Convener** is unique to each organization. The O-Convener manages registrations, maintains services, and oversees the user list for their organization, along with their private workspace.

For Data Consumers/Providers:

- **Data providers** should upload data and set up the private data service interface
- **Data consumers** can search, request, and consume data/services.

2.4 Operating Environment

E-DBA is expected to work on any operating system that can run a web browser with full functionality based on Chromium. The client machine must be capable of displaying graphical user interfaces, and network access is required.
Typically, the E-DBA server runs on a Linux distribution system. The portability of the E-DBA server software depends on the runtime environments of its components.

2.5 Design and Implementation Constraints

There are no constraints on design and implementation.

2.6 User Documentation

Each web interface page should include brief instructions for user actions. The E-DBA system provides role-specific documentation:

- **T-Admins**: Detailed guides for system setup, maintenance, and troubleshooting.
- **E-Admins/Senior E-Admins**: Workflow instructions for registration review, policy enforcement, and log auditing.
- **Conveners**: Guides on managing member lists, configuring workspace services, and checking annual fees (RMB100/person).
- **Data Providers**: Instructions for uploading data and configuring private service interfaces.
- **Data Consumers**: Tutorials on searching public course info, requesting thesis downloads, and completing payments, including batch identity verification and payment error resolution.

2.7 Assumptions and Dependencies

The successful operation of the E-DBA system relies on several key assumptions and dependencies. Key assumptions include the availability and reliability of external systems, such as student information databases and payment gateways, which must provide stable, low-latency APIs for real-time data access and transaction processing.

Critical dependencies include third-party services such as PayPal and WeChat SDKs for payment processing, SendGrid for email delivery of login codes, and cloud providers for scalable infrastructure.

E-DBA's subsystems depend on three major runtime environments:

- Apache HTTP Server (version 2.4.63)
- MySQL Community Server (version 8.4.4 LTS)
- PHP (version 8.4.4)

3. System Features

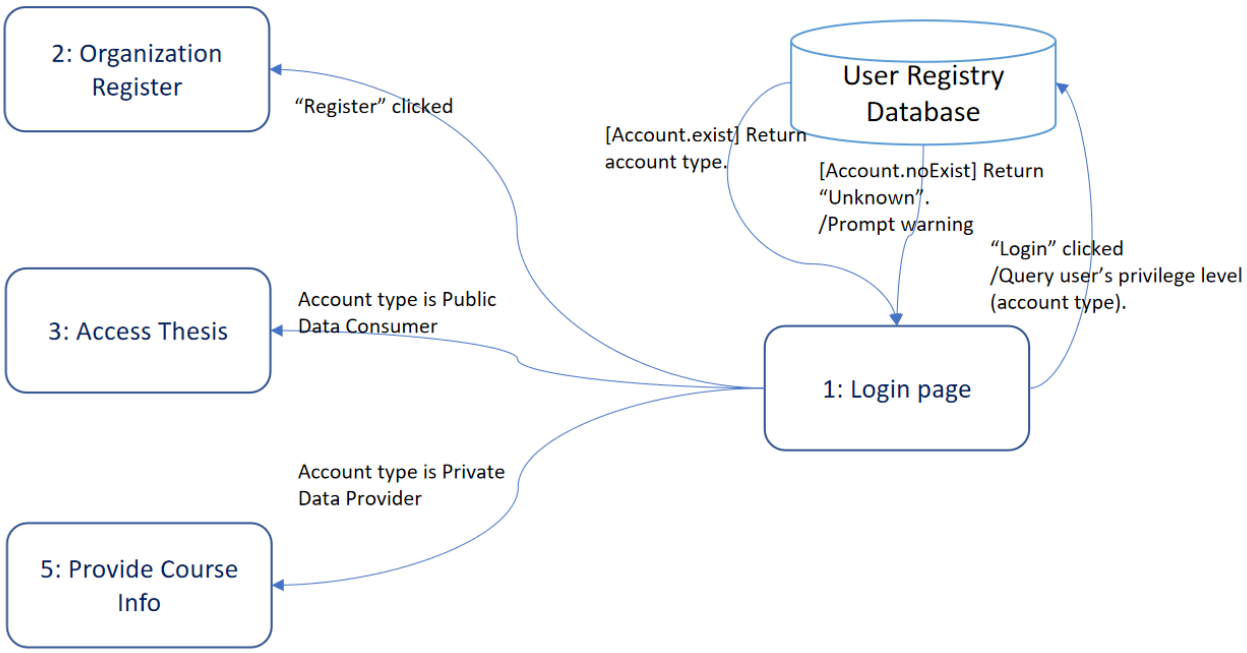
This part we will show the features of all system.

3.1 User Identification & Login

3.1.1 Description and Priority

- **Description**: All users can login with password. Login page check if the Email is registered, and query the user registry database for user's type. Since users have different privilege level according to their account type, they will be redirected to page that the user has permission to.
- **Priority**: This page is the essential portal page for any type of user to access their dedicated service page.

3.1.2 Stimulus/Response Sequences



If a registered user wants to login the system, enter their email, then click “Send Code” button. An Email with login code will be sent. Type in the code and click “Login”. The user will be automatically redirected to their corresponding page.

3.1.3 Functional Requirements

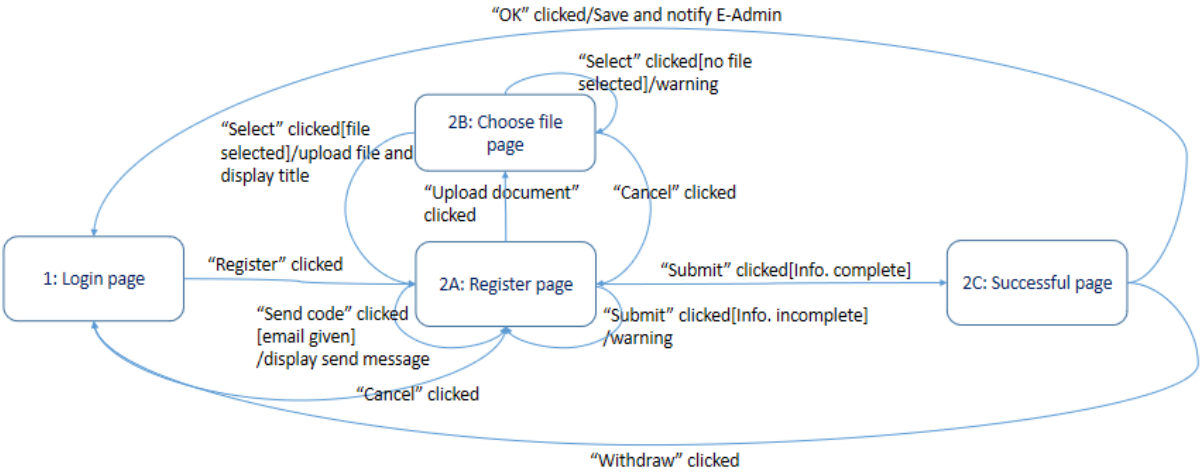
REQ-1: Cooperate with O-Convener’s register page.
REQ-2: Is able to connect to database for querying user’s account type.

3.2 Register Organization

3.2.1 Description and Priority

- **Description**: The convener registers his or her organization by providing the document to prove the convener represents the organization and their email address. If all the required info had been submitted, then E-Admin or Sr E-Admin approves or rejects the request.
- **Priority**: This is the premise for all organizations to use E-DBA.

3.2.2 Stimulus/Response Sequences



The O-Convener accesses the E-DBA system's registration entrance without logging in. Fills in personal and organization details, uploads required proof file, enters a valid email, agrees to terms, and clicks "Submit". The system verifies filled fields, file formats, and email. If there are errors, it prompts the O-Convener to correct and resubmit. After passing the initial check, the application gets a unique ID, is stored in the database, and an email is sent to confirm submission. It then waits for E-Admin's review. The E-Admin checks the application. If approved, it goes to the Senior E-Admin; if not, the O-Convener is notified with reasons. The Senior E-Admin reviews and, if satisfied, approves, and the system creates a workspace for the organization.

3.2.3 Functional Requirements

REQ-1: O-Convener can click “Register” to register his/her organization.
REQ-2: Clicking "Upload document" to upload the required file.
REQ-3: Then choose file from the file list.
REQ-4: Final O-Convener can click the “submit” to submit the application.

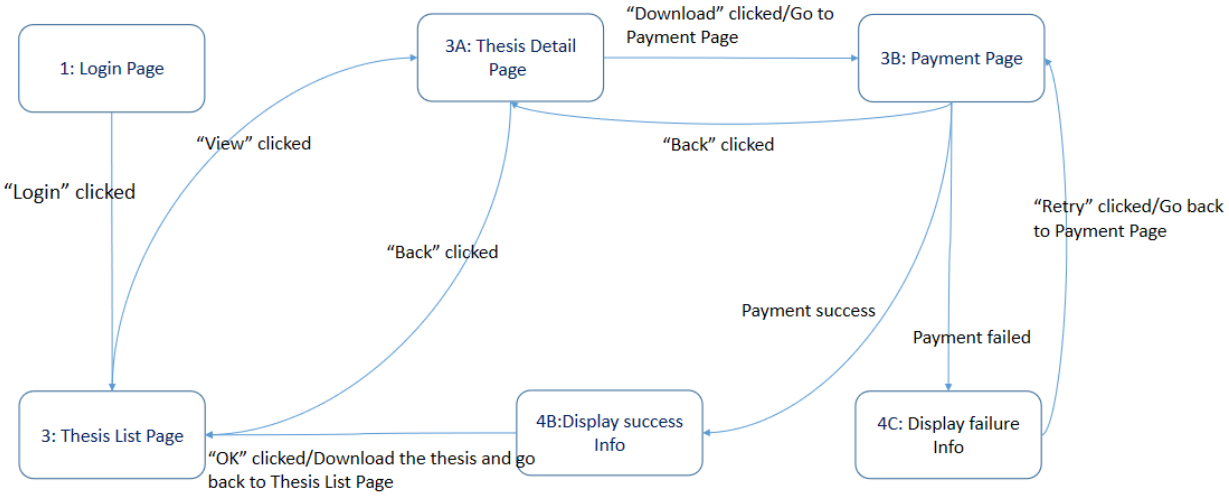
3.3 Access Thesis

3.3.1 Description and Priority

- **Description**: Actors can view all the thesis title in public data. And they can view more details about the thesis and decide whether he or she need it. Then they can pay to download the thesis to get full information.
- **Priority**: The highest priorities are implementing basic access control, bank transfer payment, and permission levels (Display/Download), as these are core features. Extended payment methods and whitelist management are lower priorities.

3.3.2

State transition diagram for Access Thesis



1. Actors can browse the list of thesis and select one. Clicking the "View" button navigates the actor to the thesis detail page.
2. Actors can view detailed information about the selected thesis. Clicking the "Download" button navigates the actor to the payment page. Clicking the "Back" button navigates the actor to the thesis list page.
3. Clicking the "Back" button navigates the actor to the thesis list page.
4. Actors can complete the payment via bank transfer. Clicking the "Back" button to return to the thesis detail page.
5. The system displays a success message and provides an option to download the thesis. Clicking the "OK" button navigates the actor to the thesis list page.
6. The system displays a failure message and provides an option to retry the payment. Clicking the "OK" button navigates the actor to the payment page.

3.3.3 Functional Requirements

REQ-1: After successful payment, actors can download the selected thesis in PDF format. Once the download is completed, the system must return the actor to the thesis list page.

REQ-2: If the payment fails, the system must display a payment failure message and provide an option to return to the payment page to retry.

REQ-3: Other requirements that pay service needs.

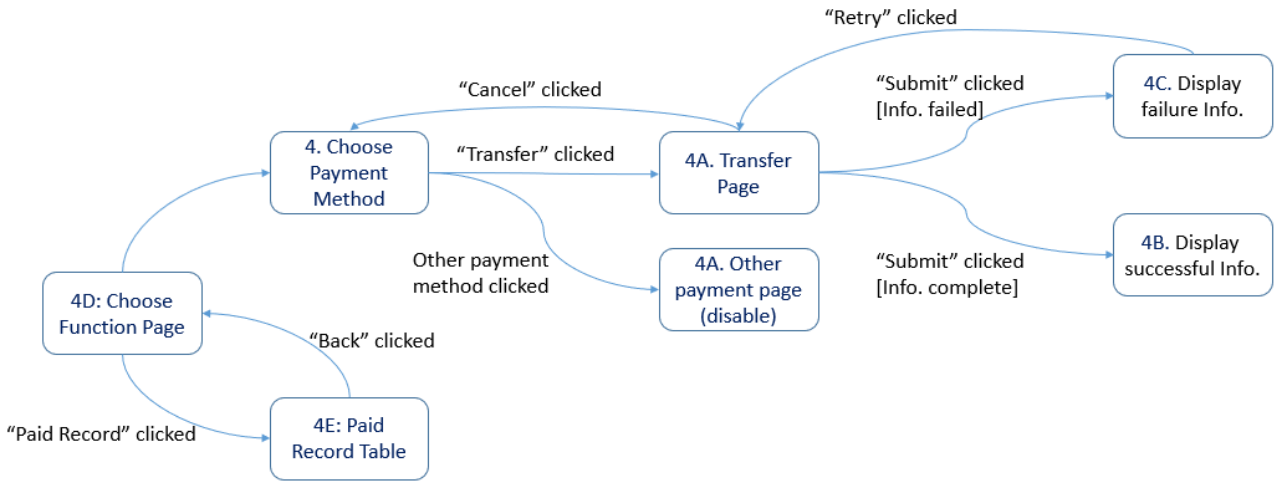
3.4 Pay Service

3.4.1 Description and Priority

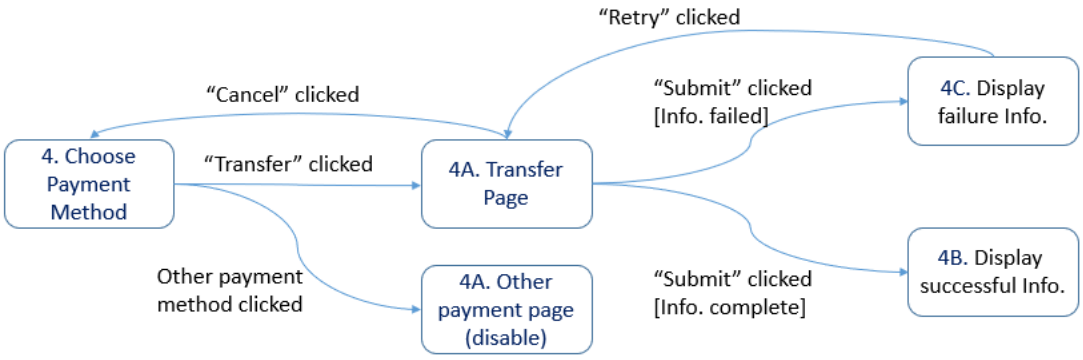
- **Description**: Actors can pay via multiple methods like PayPal, Credit Card, etc., but only transfer is allowed initially. Payment result is "success" or "fail". For instance, O-Convener may use it to pay fees for members, and in the initial stage, only transfer is available.
- **Priority**: Not explicitly stated. But as payment is crucial for system charging and operation, especially in scenarios like paying for services or accessing paid content, it has a high priority and should be focused on during development.

3.4.2 Stimulus/Response Sequences

State transition diagram for Public O-Convener



State transition diagram for Public Data Consumer



1. O – Convener can click paid record to check the table of the information include account name, account number, amount.
2. Convener and Public Data Consumer(user) enters the payment service page, sees that only the money transfer payment method is available, and clicks "Transfer" to enter the transfer page.

3. The user fills in information such as account name, bank name, account number, and password on the transfer page.
4. After filling in the information, the user clicks "Submit".
5. If the information is complete and the payment is successful, the system enters the display success information page, prompting "Payment is completed!". The user clicks "OK" to end the payment process.
6. If the information is incomplete, after clicking "Submit", the system remains on the transfer page, prompting the user to complete the information.
7. If the information is complete but the payment fails, the system enters the display failure information page, prompting "Payment is failed.". The user can click "Retry" to return to the transfer page to re - enter the information for payment, or click "Cancel" to cancel the payment and end the payment process.

3.4.3 Functional Requirements

REQ-1: After completing the payment, it will return to the page before opening the payment function.
REQ-2: Clicking "Cancel" after a failed payment will also return to the page before opening the payment function.
REQ-3: Clicking "OK" after a successful payment will also return to the page before opening the payment function.
REQ-4: Only O– Convener can view the paid record.

3.5 Provide Course Information

3.5.1 Description and Priority

This feature allows a private data provider to add new course information or edit existing course information within the system. It is High priority because managing course data is a core function that directly impacts end-users' ability to view and maintain accurate academic offerings.

3.5.2 Stimulus/Response Sequences

Navigation: The data provider login his account, accesses Course Management Page.

Form Display: The system displays the course information form.

Add course: Click the "Add" button and the go to the Add New Course Page. The data provider fills in the course details.

Edit course: Click the "Edit" button of which course you want to edit, and the go to the Edit Course Page. The data provider fills in the course details.

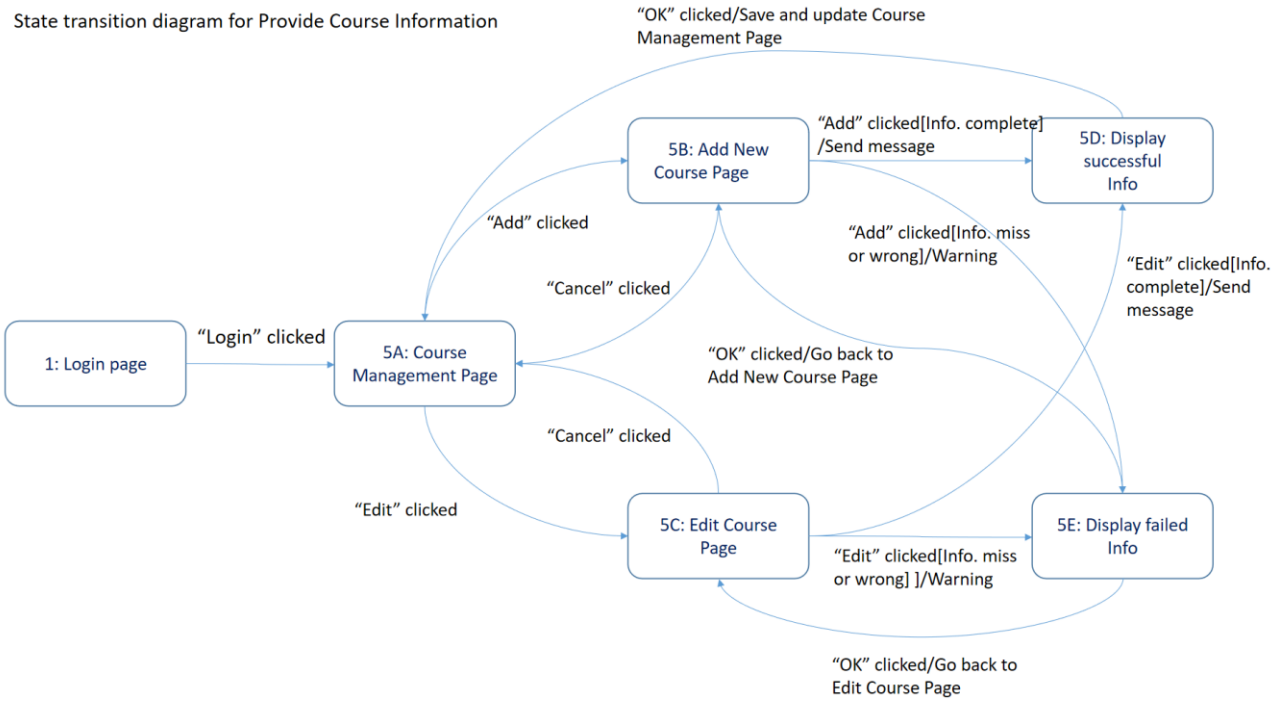
Submission: The provider submits the form.

Validation: The system validates the entered data.

Condition: If all required fields are filled and the data format is correct, validation succeeds.

Alternate: If validation fails, an error is shown.

Saving: On successful validation, the system sends a successful information and saves the new course information to the database.



3.5.3 Functional Requirements

REQ-1: The system shall provide a form for creating new courses with fields for title, description, schedule, and any other required metadata.
REQ-2: The system shall validate all mandatory fields upon submission and display an error message if any required data is missing or formatted incorrectly.
REQ-3: The system shall allow editing of existing course details (e.g., title, description, schedule) and save the changes upon successful validation.
REQ-4: If validation fails, the system shall highlight the invalid fields and display an appropriate error message.
REQ-5: The system shall confirm successful creation or update of the course with a clear confirmation message.

4. External Interface Requirements

4.1 User Interfaces

The User Interface we put in the *UI diagrams.pdf*

4.2 Hardware Interfaces

The E-DBA system interfaces with various hardware components:

- **Server-side:** High-performance servers with multi-core processors, 64GB+ RAM, SSD storage for fast access, and redundant storage (RAID or cloud) for backup. The servers are connected via high-speed internet (e.g., gigabit Ethernet or fiber).
- **Client-side:** User devices such as desktops, laptops, and mobile devices running modern browsers (Chrome, Firefox, Safari). Input devices like keyboards, mice, and touchscreens are used for interaction.
- **Data and Control Interaction:** Communication between software and hardware is managed via HTTP/HTTPS for data transmission, and APIs for user and data management. Internal communication uses TCP/IP, and sensitive data is encrypted with SSL/TLS.
- **Communication Protocols:** External system integration is supported via standard protocols (e.g., REST) for payment services like WeChat Pay, Alipay, and PayPal.

4.3 Software Interfaces

4.3.1 Database:

- **MySQL (Version 8.4.4)**: Used for storing structured data such as student information (name, ID, GPA) and thesis metadata.
- **MongoDB (Version 5.0.2):** Used for storing unstructured data like logs and transaction histories.

4.3.2 Operating Systems:

- **Server-side System:** Linux (Ubuntu 20.04 LTS):
- **Client-side System:** Windows 10 (Version 21H1) or macOS (Version 11.6)

4.3.3 Webservers:

- **Nginx (Version 1.21.0)**: Used to serve the web application and handle user requests.
- **Apache HTTP Server (Version 2.4.51)**: Alternative web server for specific configurations, offering robust handling of static and dynamic content.

4.3.4 Libraries and Tools:

- **Node.js (Version 16.5.0)**: Primary backend framework for managing API requests, handling authentication, and processing user data.
- **React (Version 17.0.2)**: Front-end framework for building interactive user interfaces, managing user inputs for data queries and system interactions.
- **Django (Version 3.2.7)**: Backend framework for developing RESTful APIs that handle student authentication, thesis sharing, and payment processing.

4.3.5 External Commercial Components:

- WeChat Pay API (Version 3.6.0)
- Alipay API (Version 3.4.0)
- PayPal API (Version 2.0.6)

4.4 Communications Interfaces

4.4.1 Communication Functions:

- **Email**: The system will utilize **SMTP** for sending email notifications regarding registration, payment confirmations, and service access (e.g., thesis access). Emails are formatted in HTML or plain text for compatibility with email clients.
- **Web Browser**: Users will interact with the system through web browsers (e.g., Google Chrome, Mozilla Firefox, Safari), which will communicate with the server over **HTTPS**. The front-end application will render the system's user interface using modern web technologies (HTML5, CSS3, JavaScript).
- **Network Server Communications**: The system will rely on **HTTP/HTTPS (TLS 1.2)** for secure client-server communication.
- **External API Communication**: The system will integrate with external APIs for payment processing (PayPal, Alipay, WeChat Pay). The communication standard will follow **RESTful** APIs, using **JSON** as the message format for ease of use and interoperability.

4.4.2 Communication Standard:

- **HTTP/HTTPS (TLS 1.2):** Users interactions with the web application will be secured using HTTPS.
- **SMTP:** Used for sending transactional emails (registration confirmations, payment receipts, etc.). Emails will adhere to standard message formats (MIME, HTML, plain text).

4.4.3 Security:

All sensitive communication, such as login credentials and payment information, will be encrypted using **SSL/TLS** to prevent interception during transmission. **OAuth 2.0** will be used for secure user authentication, ensuring that only authorized users can access specific system features like private data access or payment services. For external communications, such as payment transactions, the system will integrate with secure authorization mechanisms provided by third-party services (WeChat Pay, Alipay, PayPal).

4.4.4 Data Transfer Rates:

The system will handle varying data transfer rates depending on the operation.
For example, user authentication requests will involve minimal data transfer and should be completed within **2-3 seconds** under normal conditions. However, more data-intensive operations, such as downloading thesis PDFs or handling large file uploads, may require higher bandwidth due to the larger file sizes, and these operations may take longer to complete.

4.4.5 Synchronization Mechanisms:

Session Management: The system will use **session cookies** to manage user sessions, to ensure that once a user is authenticated, they remain logged in until they log out or their session expires.
Data Consistency: To ensure data consistency across different parts of the system (e.g., user data, transaction records), the backend will employ database **transactions**. This will ensure that no partial updates occur, maintaining data integrity during system failures.
Real-Time Updates: Where necessary, the system will utilize **WebSockets** or **Server-Sent Events (SSE)** to deliver real-time updates, such as transaction status changes or new service availability, to users.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system must support at least 1000 concurrent users during peak usage times without performance degradation. Response time for any user action should not exceed 2 seconds under normal operating conditions. For real-time services such as student identity authentication and GPA record access, the system must provide results within 1 second for 95% of requests. Also, the system must handle batch uploads of at least 10,000 entries within 5 minutes.

5.2 Safety Requirements

All data must be backed up daily to prevent loss in case of system failure. The system must include safeguards to prevent unauthorized access to sensitive data, such as student records and financial transactions. In case of a system crash, the system must recover within 10 minutes without data corruption. Additionally, the system must comply with relevant safety regulations and certifications, such as data sovereignty laws and corporate polices.

5.3 Security Requirements

All user data must be encrypted using AES-256 encryption. User authentication must include two-factor authentication for all roles, including administrators, data providers, and consumers. Access to private data must be restricted based on user roles and permissions as defined by the O-Convener. The system must log all user activities for auditing purposes. Logs must be immutable and accessible only to authorized administrators. The system must comply with external security regulations, such as GDPR or equivalent local laws, to ensure data privacy and protection.

5.4 Software Quality Attributes

Adaptability: The system should support the addition of new services with minimal changes to the existing architecture.
Availability: The system must maintain an uptime of 99.9%, ensuring continuous access of users.
Maintainability: The system should allow administrator to update configurations without requiring downtime.
Interoperability: The system must integrate seamlessly with external databases through configurable interfaces.
Scalability: The system should be able to scale to support additional organizations and users without significant performance degradation.
Usability: The user interface must be intuitive and provide clear instructions for all user roles, including O-Conveners, data providers, and consumers.
Testability: All system components must be testable, with automated tests covering 80% of the codebase.

6. Other Requirements

Appendix A: Glossary

Abbreviation	Full Explanation
T-Admin	Technical Administrator
E-Admin	Management admin
O-Convener	Organization Convener

Appendix B: Analysis Models

<To be implemented>

Appendix C: Issues List

<To be implemented>